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**“MODELING AND SIMULATION OF A SATELLITE
TRAINING KIT FOR PROVIDING SPACE SYSTEMS
ENGINEERING KNOWLEDGE AND TECHNOLOGICAL
DEVELOPMENT IN STEAM”**

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INTRODUCTION

- Aerospace Engineering is a part of modern engineering.
- Limited knowledge & resource.
- Big satellites are costly and complex to build.

THE AIM BEHIND SATELLITE KIT

- ❖ To increase interest among the students and people-in-general.
- ❖ To provide learning opportunity.
- ❖ Inspiring students to get involved in space related research.
- ❖ Providing knowledge through assembly, integration and operation of a satellite training kit.

SUB-SYSTEMS OF THE KIT

1. Chassis
2. Electric Power System (EPS)
3. On Board Computer (OBC)
4. Communication System (COM)
5. Ground Station (GS)
6. Payloads

OVERALL BLOCK DIAGRAM

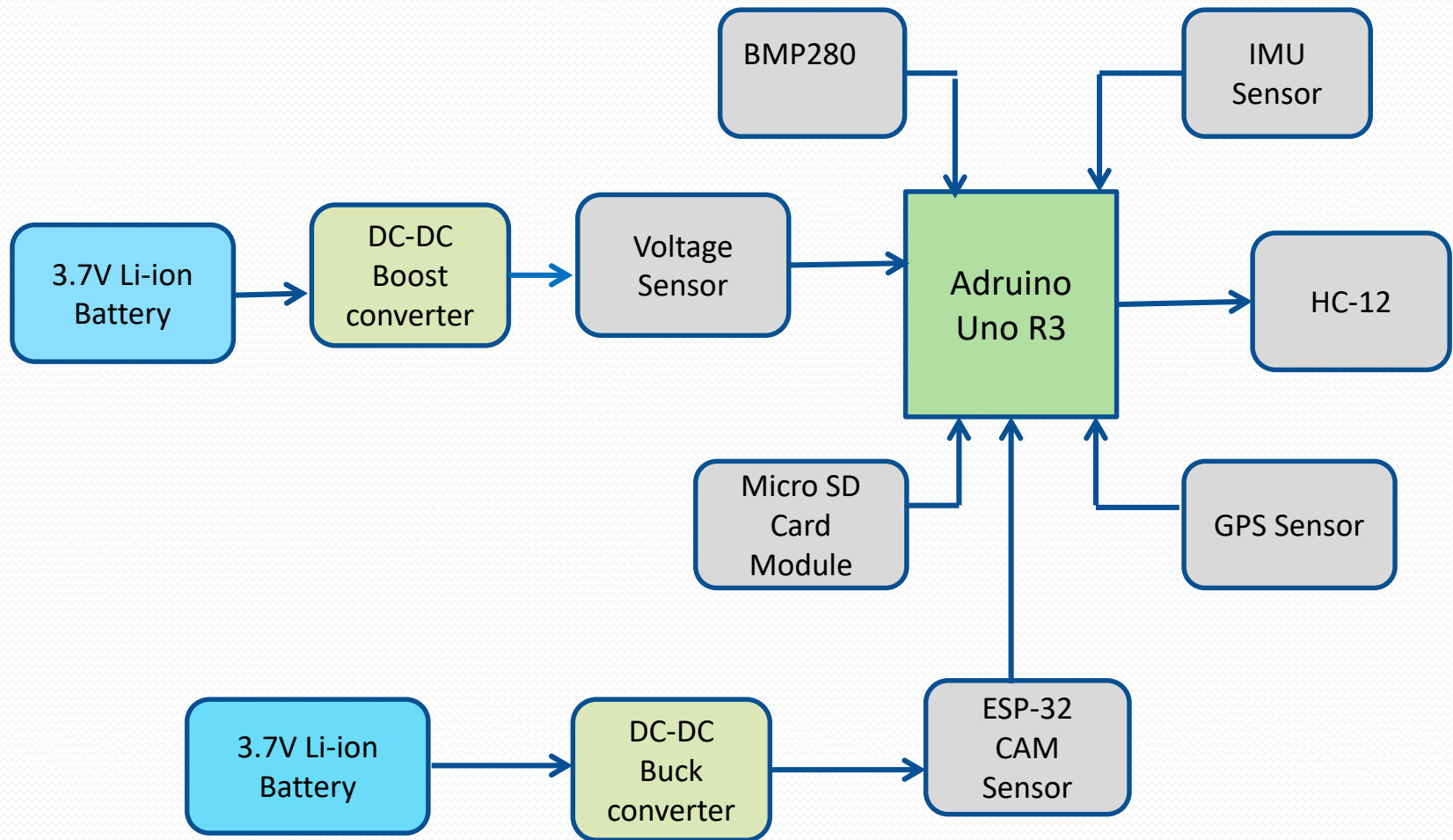
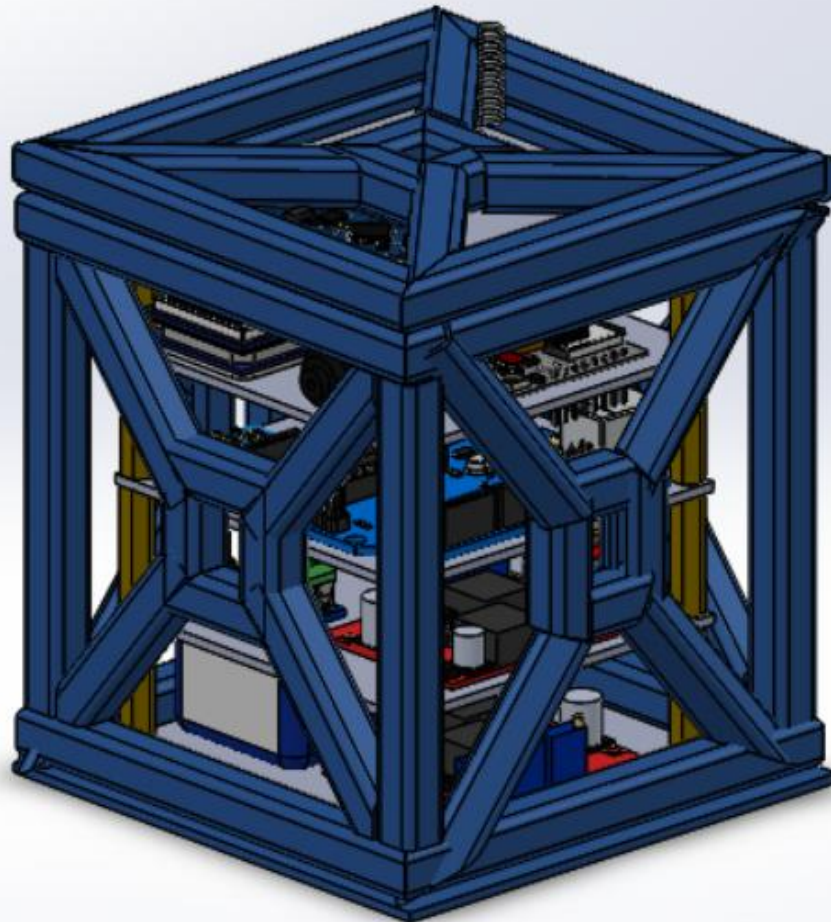
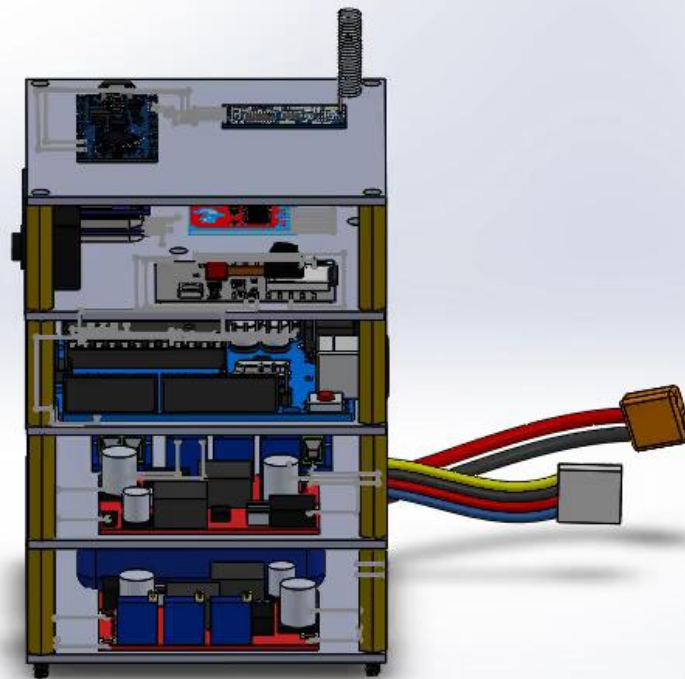


Fig: Overall Block diagram

MECHANICAL VIEW OF THE KIT



SPLIT-VIEW OF THE STACK



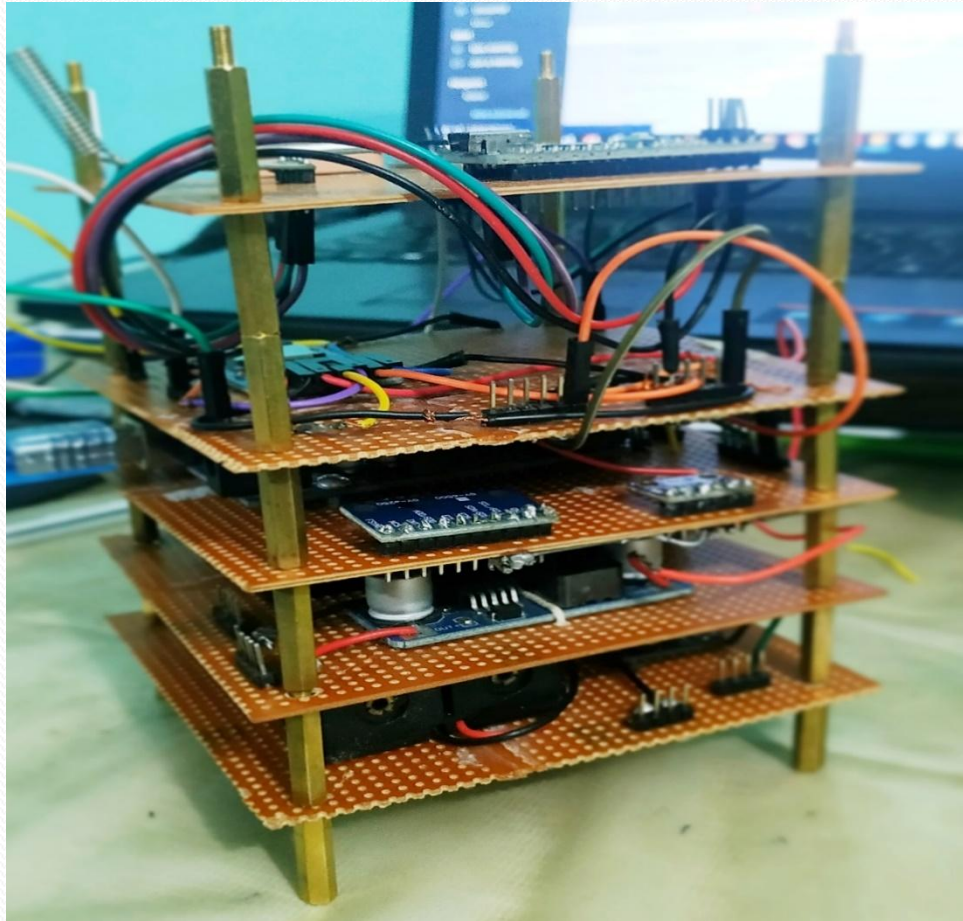
SPLIT-VIEW OF THE KIT





CHASSIS

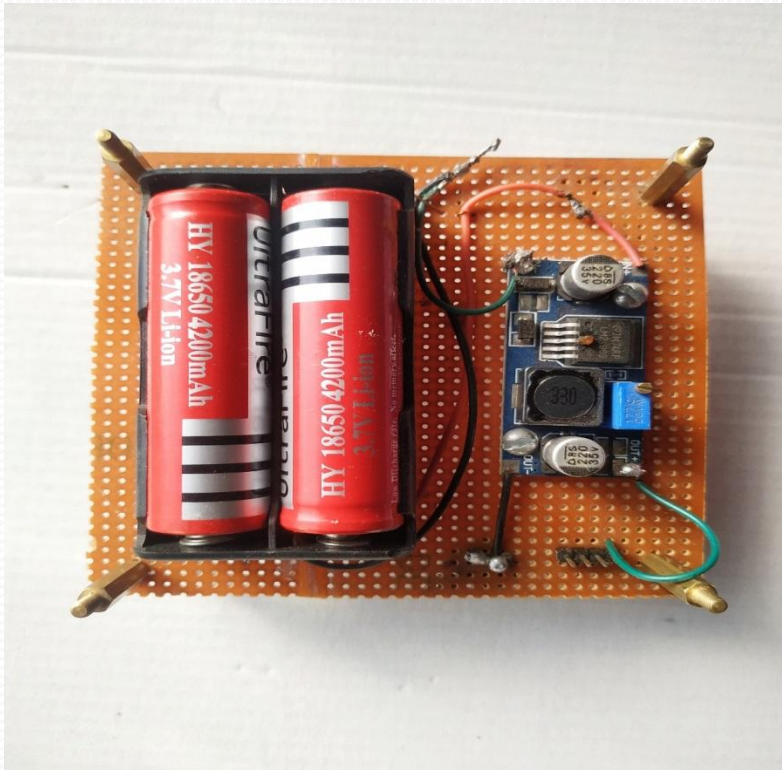
INNER STACK



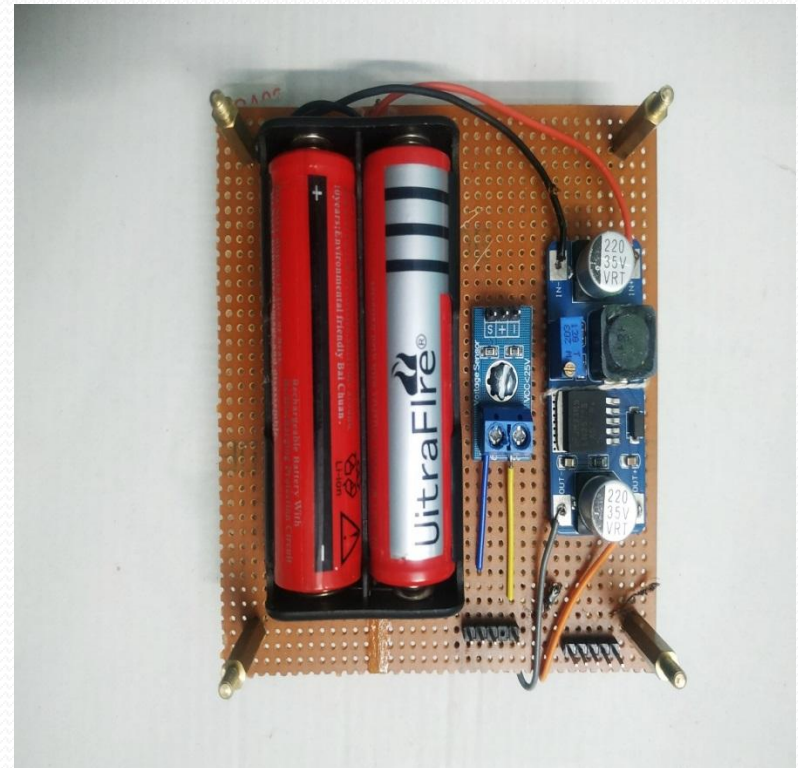


ELECTRICAL POWER SYSTEM (EPS)

EPS CIRCUIT



EPS 1st Stack

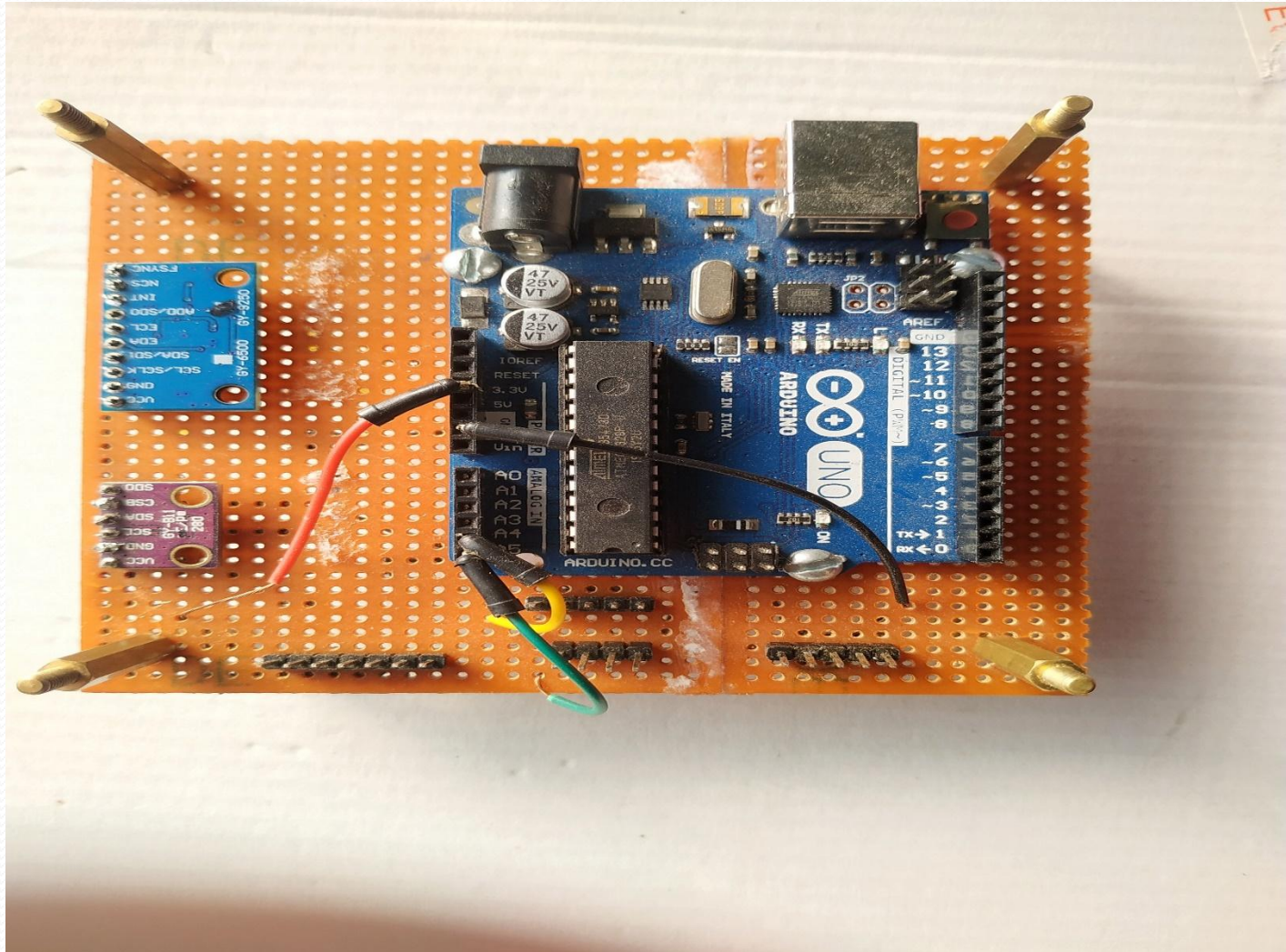


EPS 2nd Stack

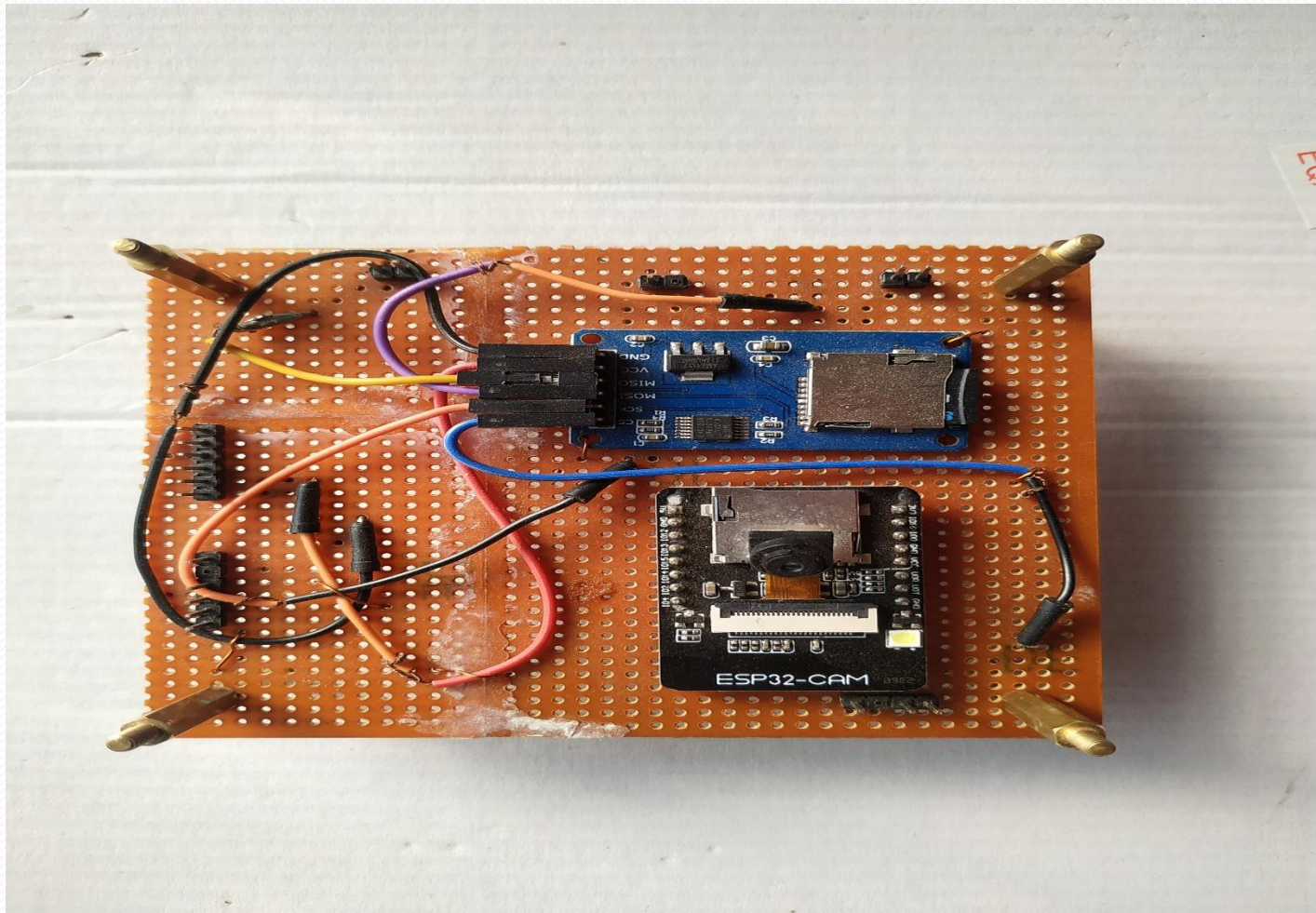


ON-BOARD COMPUTER SYSTEM (OBC)

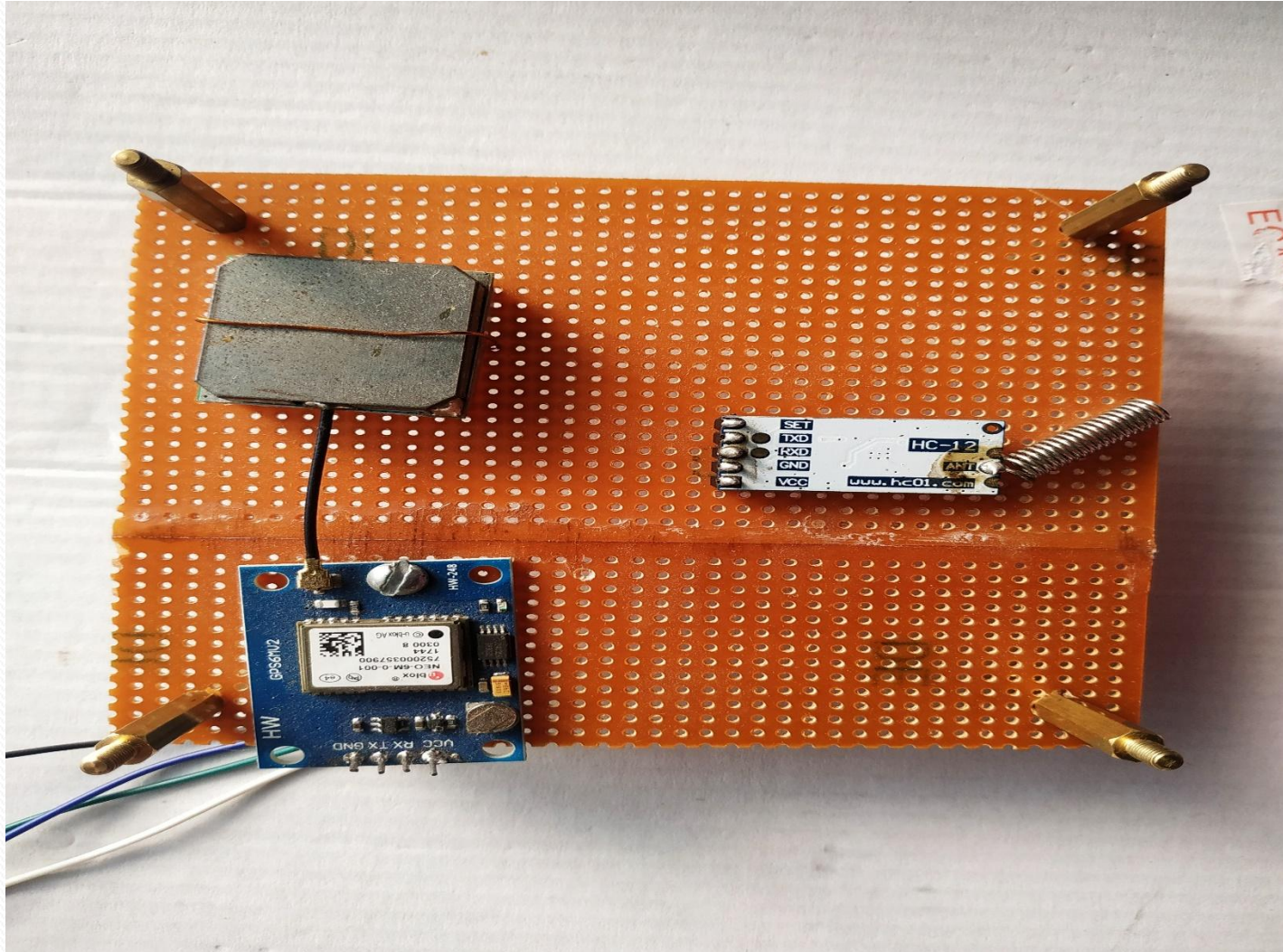
OBC STACK-1



OBC STACK-2



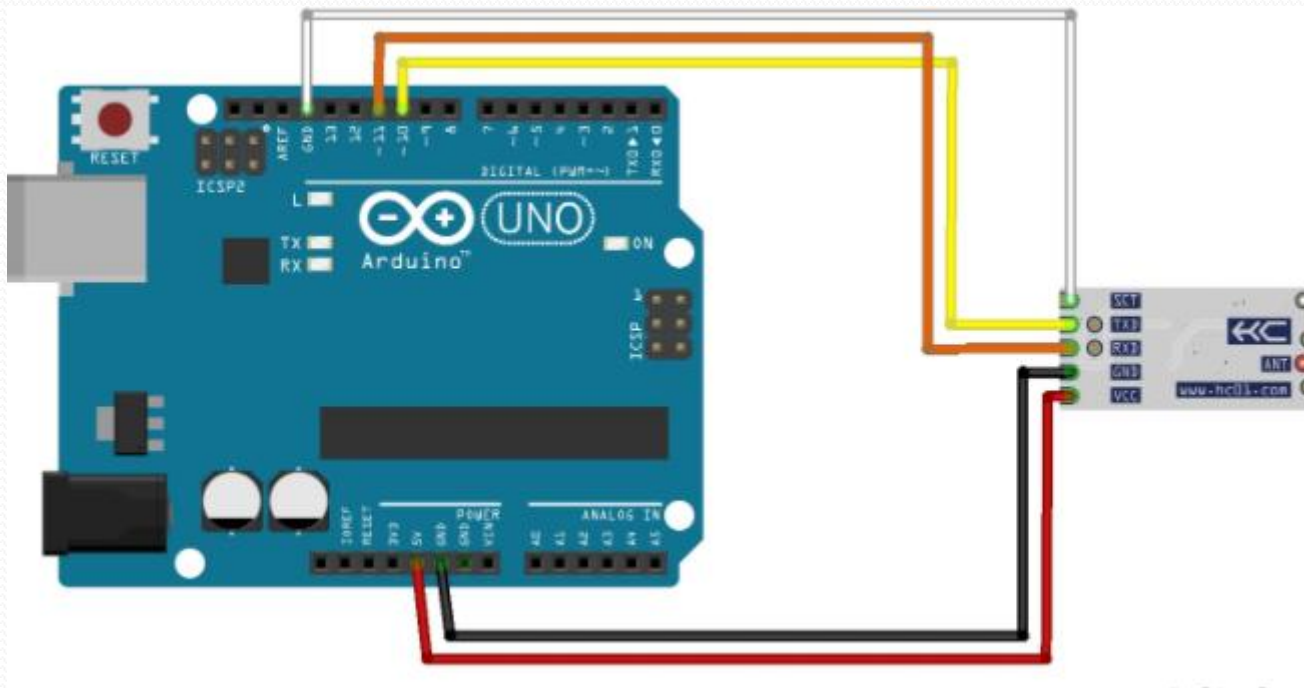
COM STACK



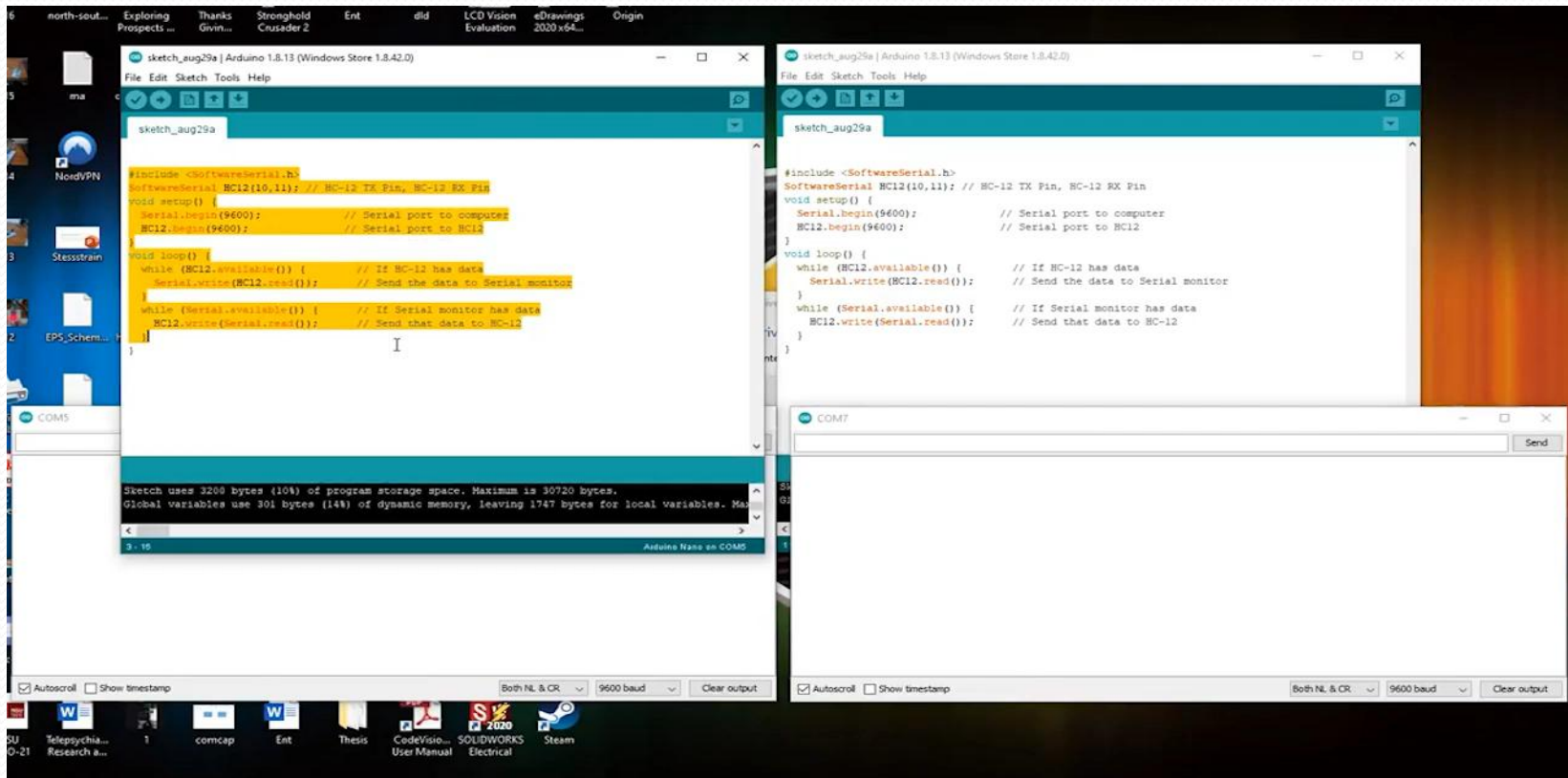


COMMUNICATION SYSTEM (COM)

COMMUNICATION DIAGRAM



DATA TRANSFER VIA HC-12

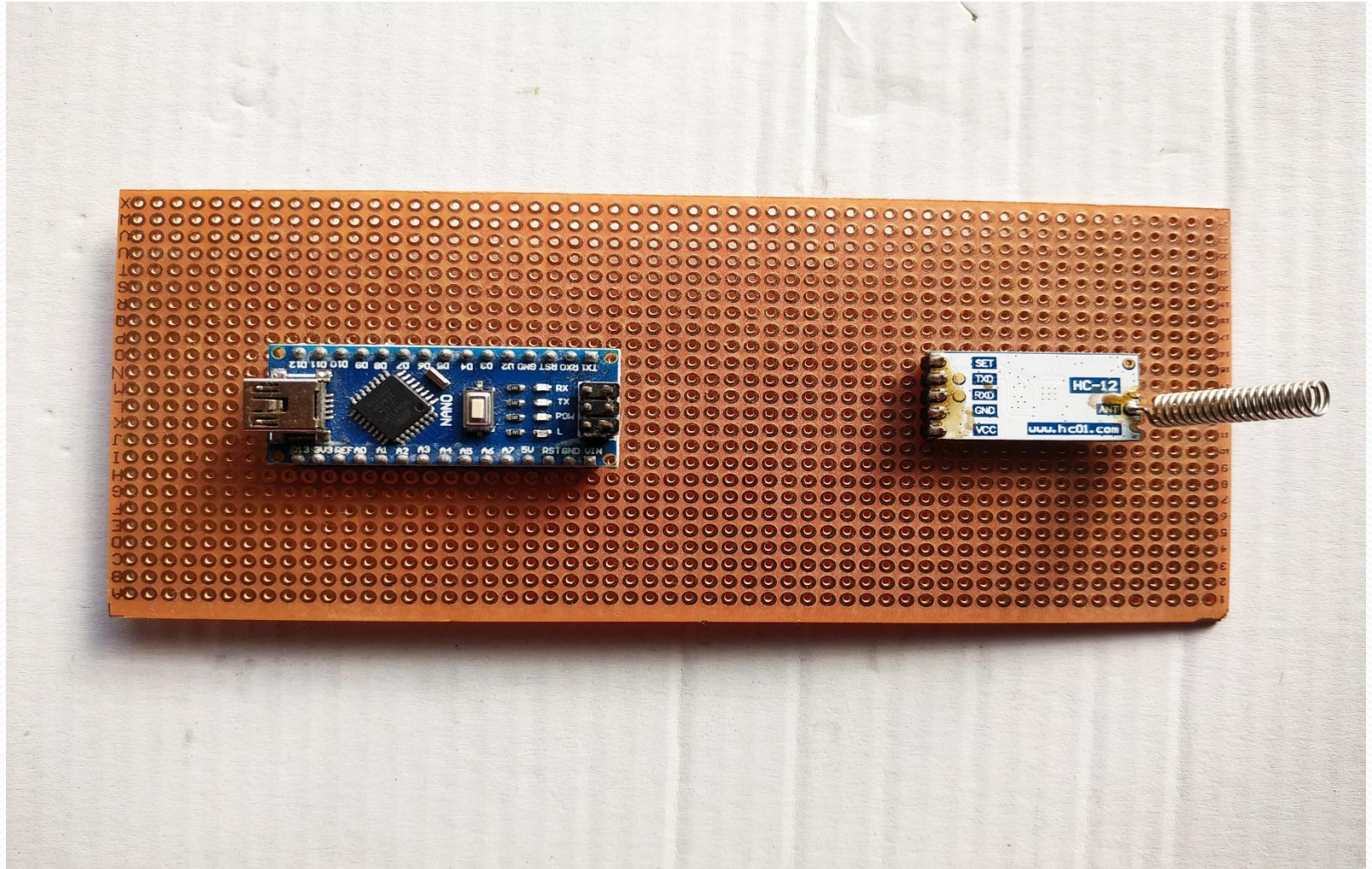




GROUND STATION

(GS)

GROUND STATION CIRCUIT



RESULTS

```
COM3
14:30:19.851 -> magZ: 60.87
14:30:19.885 -> horizontal direction: 54.82
14:30:19.885 -> at 6935ms
14:30:19.920 ->
14:30:21.397 -> sensorId: 115
14:30:21.397 -> accelX: -0.06
14:30:21.432 -> accelY: 0.02
14:30:21.432 -> accelZ: -1.05
14:30:21.467 -> accelSqrt: 1.05
14:30:21.467 -> gyroX: 1.16
14:30:21.467 -> gyroY: 0.06
14:30:21.502 -> gyroZ: 0.43
14:30:21.502 -> magX: 71.48
14:30:21.536 -> maxY: 52.42
14:30:21.536 -> magZ: 60.87
14:30:21.536 -> horizontal direction: 53.75
14:30:21.569 -> at 8614ms
14:30:21.604 ->
14:30:23.061 -> sensorId: 115
14:30:23.095 -> accelX: -0.07
14:30:23.095 -> accelY: 0.03
14:30:23.131 -> accelZ: -1.02
14:30:23.131 -> accelSqrt: 1.03
14:30:23.162 -> gyroX: 0.67
14:30:23.162 -> gyroY: -0.12
14:30:23.162 -> gyroZ: 0.55
14:30:23.197 -> magX: 72.68
14:30:23.197 -> maxY: 51.23
14:30:23.230 -> magZ: 63.16
14:30:23.230 -> horizontal direction: 54.82
14:30:23.264 -> at 10296ms
14:30:23.264 ->
14:30:24.749 -> sensorId: 115
14:30:24.784 -> accelX: -0.06
14:30:24.784 -> accelY: 0.03
14:30:24.784 -> accelZ: -1.04
```

Activate Windows
Go to Settings to activate Windows.

☐ Autoscroll ☒ Show timestamp

Newline 9600 baud Clear output

Result for IMU Sensor Data

RESULTS

FullExample_GPS | Arduino 1.8.12
File Edit Sketch Tools Help

FullExample_GPS

```
// The TinyGPS++ object
TinyGPSPlus gps;

// The serial connection to the GPS device
SoftwareSerial ss(RXPin, TXPin);

void setup()
{
  Serial.begin(115200);
  ss.begin(GPSBaud);

  Serial.println(F("FullExample.ino"));
  Serial.println(F("An extensive example of many interesting things"));
  Serial.print(F("Testing TinyGPS++ library v. "));
  Serial.println(TinyGPSPlus::libraryVersion());
  Serial.println(F("by Mikal Hart"));
  Serial.println();
  Serial.println(F("Sats HDOP Latitude Longitude Fix Date Time Date Alt Course Speed Card Distance Course Car"));
  Serial.println(F("----- (deg) (deg) Age Age (m) --- from GPS ---- to London ---"));
  Serial.println(F("-----"));

  void loop()
  {
    static const double LONDON_LAT = 51.508131, LONDON_LO...
```

COM3

	Sats	HDOP	Latitude (deg)	Longitude (deg)	Fix Age	Date	Time	Date Age	Alt (m)	Course ---	Speed from GPS	Card ----	Distance to London	Course to London	Car ---
5	3.1	23.444496	91.141181	441	04/10/2021	14:44:03	557	10.90	0.00	0.20	N	8076	319.31	NW	
5	3.1	23.444496	91.141181	451	04/10/2021	14:44:04	565	11.00	0.00	0.07	N	8076	319.31	NW	
5	3.1	23.444496	91.141181	459	04/10/2021	14:44:05	576	11.10	0.00	0.09	N	8076	319.31	NW	
5	3.1	23.444496	91.141181	470	04/10/2021	14:44:06	585	11.20	0.00	0.07	N	8076	319.31	NW	
5	3.1	23.444494	91.141181	479	04/10/2021	14:44:07	594	11.30	0.00	0.09	N	8076	319.31	NW	
5	3.1	23.444494	91.141181	488	04/10/2021	14:44:08	603	11.40	0.00	0.11	N	8076	319.31	NW	
5	3.1	23.444494	91.141181	497	04/10/2021	14:44:09	612	11.50	0.00	0.15	N	8076	319.31	NW	
5	3.1	23.444494	91.141181	507	04/10/2021	14:44:10	622	11.50	0.00	0.13	N	8076	319.31	NW	
5	3.1	23.444494	91.141181	516	04/10/2021	14:44:11	632	11.60	0.00	0.04	N	8076	319.31	NW	
5	3.1	23.444496	91.141189	525	04/10/2021	14:44:12	641	11.50	0.00	0.07	N	8076	319.31	NW	
5	3.1	23.444496	91.141189	532	04/10/2021	14:44:13	647	11.60	0.00	0.31	N	8076	319.31	NW	
5	3.1	23.444496	91.141189	544	04/10/2021	14:44:14	659	11.60	0.00	0.33	N	8076	319.31	NW	
5	3.1	23.444496	91.141189	554	04/10/2021	14:44:15	669	11.60	0.00	0.28	N	8076	319.31	NW	
5	3.1	23.444496	91.141189	563	04/10/2021	14:44:16	678	11.70	0.00	0.20	N	8076	319.31	NW	
5	3.1	23.444496	91.141189	573	04/10/2021	14:44:17	688	11.70	0.00	0.15	N	8076	319.31	NW	
5	3.1	23.444496	91.141189	580	04/10/2021	14:44:18	696	11.70	0.00	0.17	N	8076	319.31	NW	
5	3.1	23.444496	91.141189	592	04/10/2021	14:44:19	706	11.80	0.00	0.09	N	8076	319.31	NW	
5	3.1	23.444496	91.141189	600	04/10/2021	14:44:20	716	11.80	0.00	0.11	N	8076	319.31	NW	
5	3.1	23.444496	91.141189	610	04/10/2021	14:44:21	725	11.70	0.00	0.04	N	8076	319.31	NW	
5	3.1	23.444496	91.141189	618	04/10/2021	14:44:22	734	11.80	0.00	0.11	N	8076	319.31	NW	
5	3.1	23.444494	91.141189	628	04/10/2021	14:44:23	743	11.90	0.00	0.19	N	8076	319.31	NW	
5	3.1	23.444496	91.141189	636	04/10/2021	14:44:24	751	11.90	0.00	0.11	N	8076	319.31	NW	
5	3.1	23.444496	91.141189	648	04/10/2021	14:44:25	763	12.00	0.00	0.13	N	8076	319.31	NW	

☐ Autoscroll ☐ Show timestamp

Newline 115200 baud Clear output

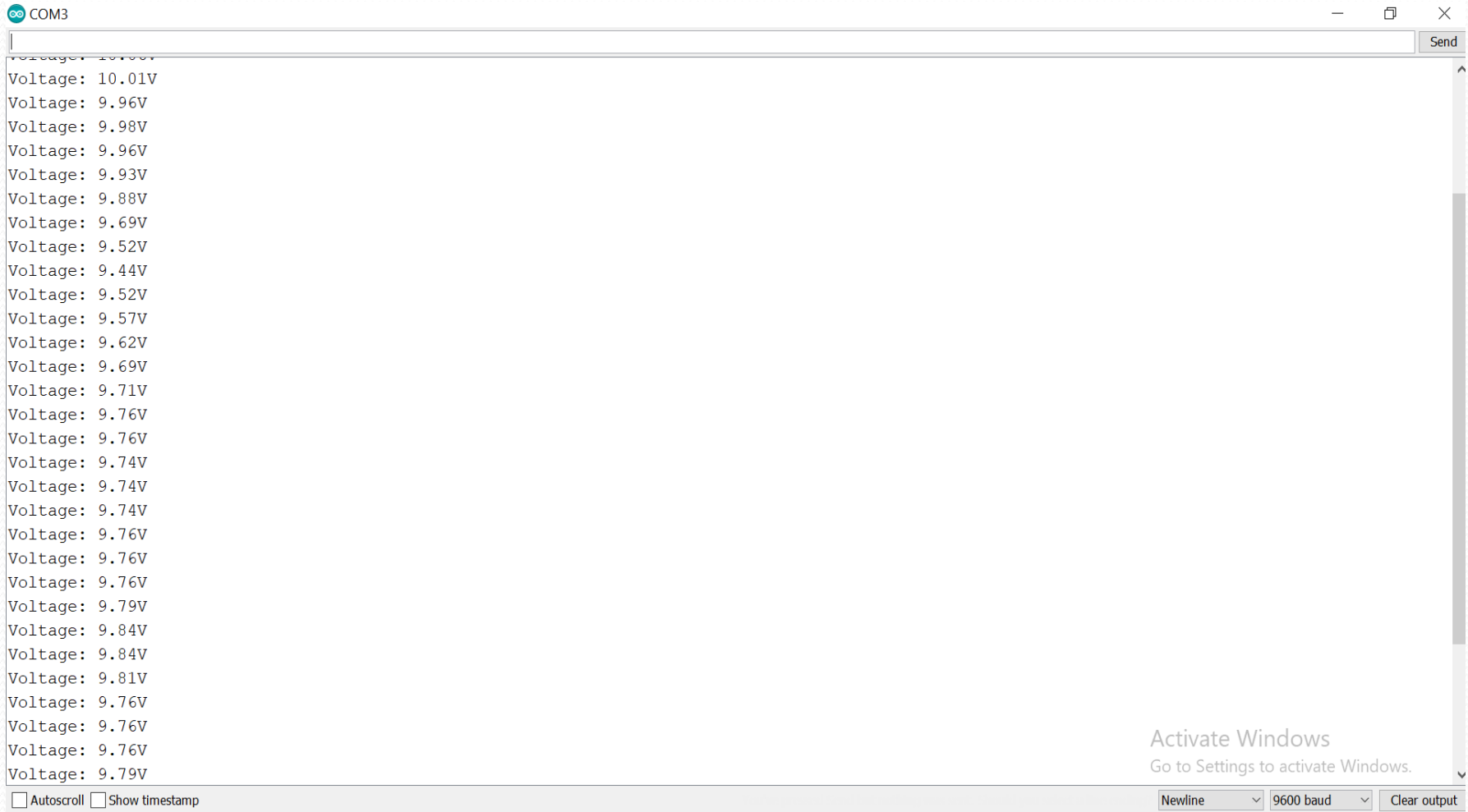
Global variables use 660 bytes (32%) of dynamic memory, leaving 1388 bytes for local variables. Maximum is 2048 bytes.

Activate Windows
Go to Settings to activate Windows

1 Arduino Uno on COM3

Result for GPS Sensor

RESULTS



A screenshot of a terminal window titled 'COM3'. The window displays a list of voltage readings from a sensor. The readings start at 10.01V and then fluctuate between approximately 9.44V and 9.84V. The terminal has a 'Send' button in the top right corner. At the bottom, there are checkboxes for 'Autoscroll' and 'Show timestamp', and a status bar showing 'Newline', '9600 baud', and 'Clear output'.

```
COM3
Voltage: 10.01V
Voltage: 9.96V
Voltage: 9.98V
Voltage: 9.96V
Voltage: 9.93V
Voltage: 9.88V
Voltage: 9.69V
Voltage: 9.52V
Voltage: 9.44V
Voltage: 9.52V
Voltage: 9.57V
Voltage: 9.62V
Voltage: 9.69V
Voltage: 9.71V
Voltage: 9.76V
Voltage: 9.76V
Voltage: 9.74V
Voltage: 9.74V
Voltage: 9.74V
Voltage: 9.76V
Voltage: 9.76V
Voltage: 9.76V
Voltage: 9.79V
Voltage: 9.84V
Voltage: 9.84V
Voltage: 9.81V
Voltage: 9.76V
Voltage: 9.76V
Voltage: 9.76V
Voltage: 9.79V
```

Activate Windows
Go to Settings to activate Windows.

☐ Autoscroll ☐ Show timestamp

Newline 9600 baud Clear output

Result for Voltage sensor

RESULTS

BMP280LOG - Notepad

File Edit Format View Help

00

Pressure (64 bit): 101076.39

Temperature: 28.08

Humidity: 73.59

00

Pressure (64 bit): 101074.21

Temperature: 28.08

Humidity: 73.61

00

Pressure (64 bit): 101073.65

Temperature: 28.07

Humidity: 73.60

00

Pressure (64 bit): 101077.80

Temperature: 28.04

Humidity: 73.61

00

Pressure (64 bit): 101076.13

Temperature: 28.04

Humidity: 73.63

00

Pressure (64 bit): 101079.07

Temperature: 28.03

Humidity: 73.65

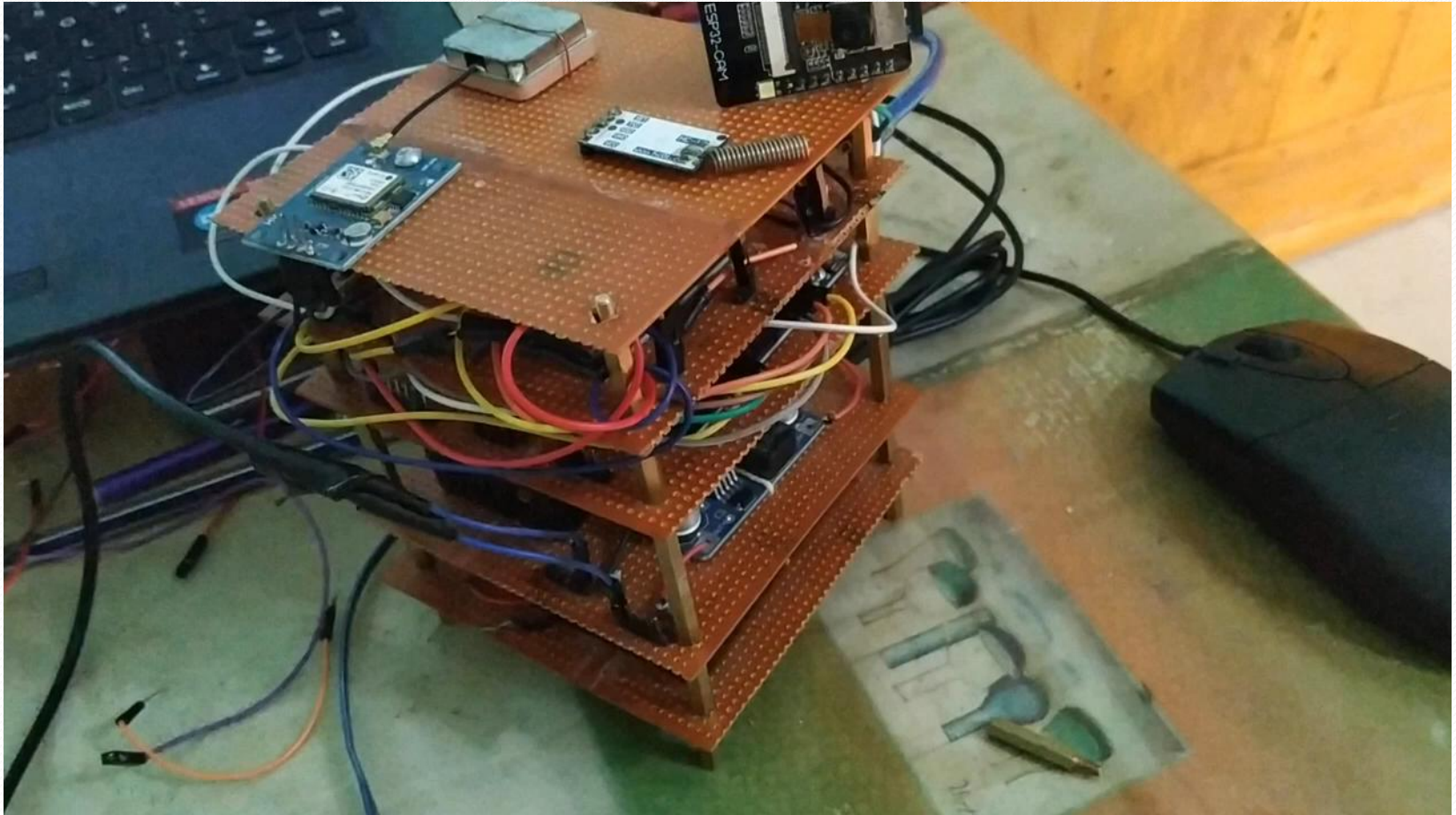
Activate Windows

Go to Settings to activate Windows.

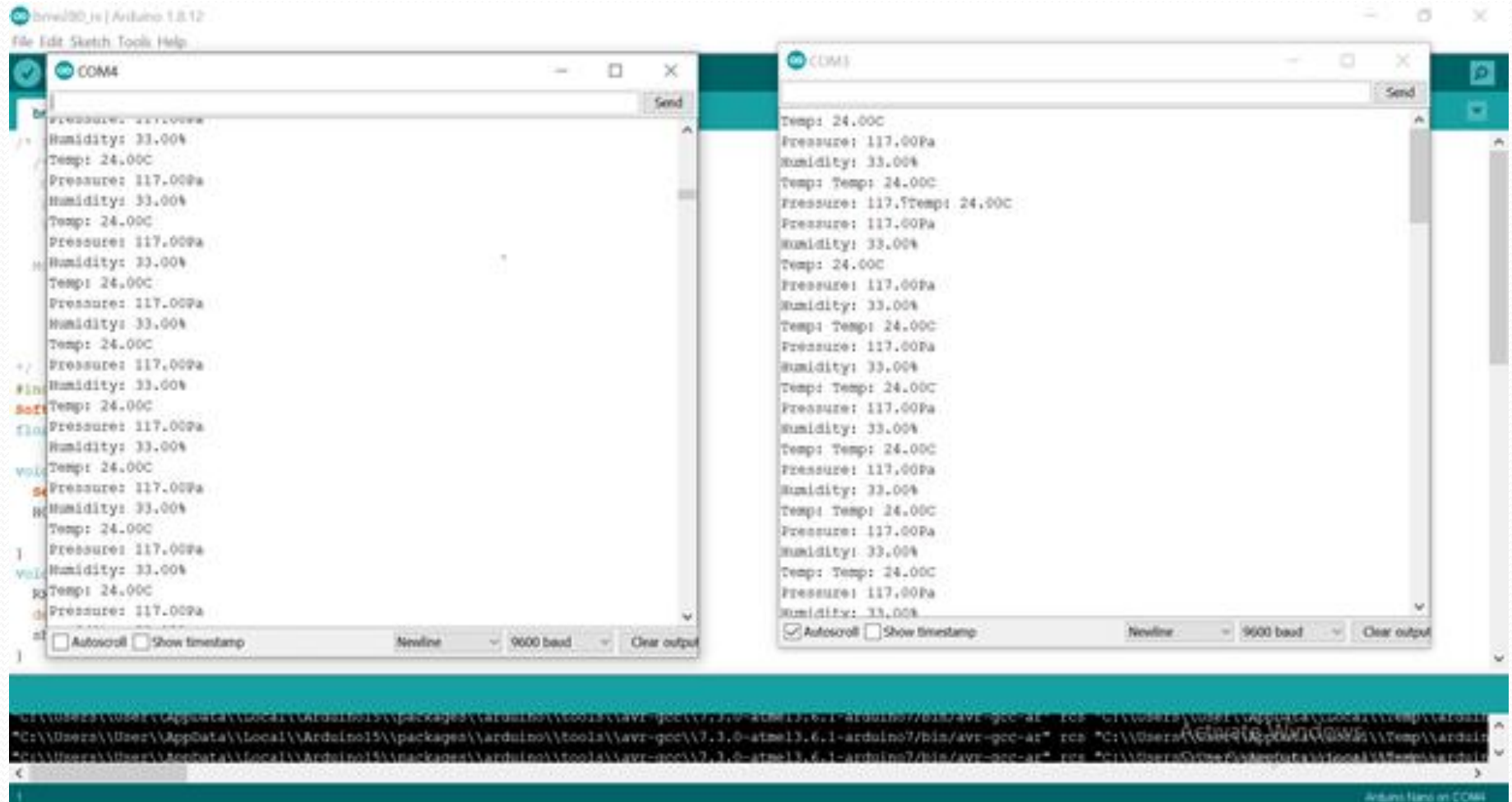
Ln 1, Col 1 100% Windows (CRLF) ANSI

Data logging of the Micro-SD card module

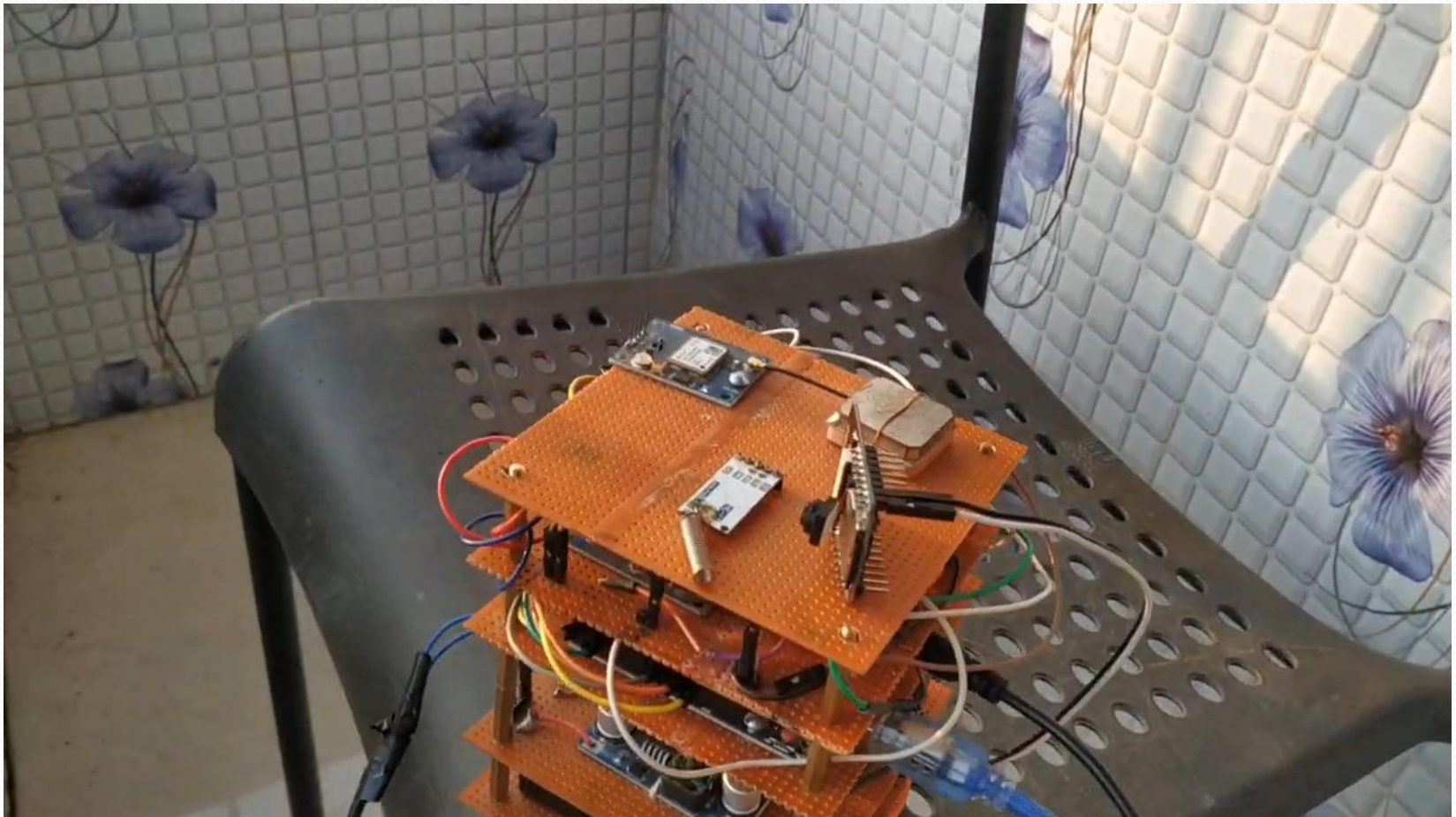
SHORT RANGE DEMONSTRATION



Result for Short range communication



LONG RANGE DEMONSTRATION



RESULTS

The screenshot displays the Arduino IDE interface. The top toolbar includes icons for File, Edit, Sketch, Tools, and Help, along with a 'Save' button. The sketch is named 'bme280_rx'. The code in the editor is as follows:

```
#include <SoftwareSerial.h>
SoftwareSerial HC12(10, 11); // HC-12 TX Pin, HC-12 RX Pin
float rxInt[5];

void setup() {
  Serial.begin(9600);          // Serial port to computer
  HC12.begin(9600);           // Serial port to HC12
}

void loop() {
  RXData();
  delay(100);
  showdata();
}

void RXData(){
  static byte ndx = 0;
  char startMarker = '<';
  char endMarker = '>';
  float rc;
  while (HC12.available() > 0) {
    rc = HC12.read();
    if (rc != endMarker) {
      rxInt[ndx] = rc;
      ndx++;
      if (ndx >= 5) {
        ndx = 5 - 1;
      }
    }
  }
}
```

The serial monitor, titled 'COM3', shows the following output:

```
Humidity: 65.00%
Temp: 31.00C
Pressure: 2.00Pa
Humidity: 64.00%
Temp: 31.00C
Pressure: 2.00Pa
Humidity: 64.00%
Temp: Temp: 31.00C
Pressure: 8.00Pa
Humidity: 65.00%
Temp: 31.00C
Pressure: 7.00Pa
Humidity: 65.00%
Temp: 31.00C
Pressure: 10.00Pa
Humidity: 64.00%
Temp: 31.00C
Pressure: 10.00Pa
Humidity: 64.00%
Temp: 31.00C
Pressure: 11.00Pa
Humidity: 64.00%
Temp: 31.00C
Pressure: 10.00Pa
Humidity: 64.00%
Temp: 31.00C
Pressure: 8.00Pa
```

At the bottom of the IDE, a status bar indicates: 'C:\Users\User\AppData\Local\Arduino15\packages\arduino\tools\avr-gcc\7.3.0-atmel3.6.1-arduino7/bin/avr-size' -A 'C:\Users\User\AppData\Local\Temp\arduino_build_859290/bme280'. Below this, it states: 'Sketch uses 5442 bytes (16%) of program storage space. Maximum is 32256 bytes. Global variables use 374 bytes (18%) of dynamic memory, leaving 1674 bytes for local variables. Maximum is 2048 bytes.'

Result for long range communication

FUTURE TASK

- ☐ Campaigning
- ☐ Training Methodology
- ☐ Assemble & Disassemble
- ☐ Further Development

CONCLUSION

- ❖ To sum up, our satellite training is an education with the aim of experiencing the development process of nanosat in a short time and acquiring the basic knowledge of space technology
- ❖ We believe that these developments will promote public and student interest in space engineering and related fields
- ❖ We hope that our work can make an impact during and after Covid-19 pandemic



THANK YOU